

Trace tables

Revise 360

Worksheet for the 360° experience: Trace table observatory

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Trace table observatory. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Complete trace tables.
- Trace selection and nested iteration.
- Use traces to locate logic errors.

Starter: before you put the headset on

$x = 2$, $x = x + 3$, `print(x)`. List each value stored in x and the output.

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Key terminology

Trace table

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Dry run

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The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Record each change

Turn right

Key facts in your own words:

Example to trace:

```
x = 2
y = x + 3
x = y * 2
print(x)
```

Challenge: What belongs in the output column?

Trace the example above. Record changes and outputs in execution order.

Step	x	y	Output

Does an assignment automatically produce output?

A trace table records output...

2 Use current values

Turn right

Key facts in your own words:

Example to trace:

a = 3
b = a + 2
a = b + 4
b = a - 1

Challenge: What is b after its final assignment?

Trace the example above. Record changes and outputs in execution order.

Step	a	b	Output

x = 4; x = x * 2; y = x + 1. What is y?

a = 3; b = a + 2; a = b + 4. Final a?

3 Trace only the chosen branch

Turn around

Key facts in your own words:

Example to trace:

```
x = 7
if x >= 5 then
  y = x + 1
else: y = 0
print(y)
```

Challenge: Which assignment is skipped?

Trace the example above. Record changes and outputs in execution order.

Step	x	y	Output

For x = 3 in the displayed algorithm, what is printed?

A skipped ELSE assignment should...

4 Trace a running total

Turn around

Key facts in your own words:

Example to trace:

```
total = 0
for i = 1 to 3
  total = total + i
next i
print(total)
```

Challenge: What are the totals after iterations 1, 2 and 3?

Trace the example above. Record changes and outputs in execution order.

Step	i / input	total	Output

Same algorithm but i = 1 to 4. Final total?

Totals after adding 1, then 2, then 3 from zero?

5 Trace nested loops

Turn left

Key facts in your own words:

Example to trace:

```
for a = 1 to 2
  for b = 1 to 2
    print(a + b)
  next b
next a
```

Challenge: List the outputs in execution order.

Trace the example above. Record changes and outputs in execution order.

Step	a	b	Output

How many output events occur here?

Outer a = 1..2, inner b = 1..2, print a+b. First output?

6 Locate a fault with a trace

Turn left

Key facts in your own words:

Example to trace:

total = 0

Inputs: 3, 4, 5

Wrong update: total = value

Challenge: Which update produces totals 3, 7 and 12?

Trace the example above. Record changes and outputs in execution order.

Step	i / input	total	Output

Why is tracing useful for debugging?

total = value differs from total = total + value because it...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

total = 0. For i = 1 to 3, add i * 2. What is printed after the loop?

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n = 4. While n > 1, print n then subtract 1. Outputs?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 14

Key question

How can you check what a program does without running it?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Trace $x = 5$, $y = x - 2$, $x = y * 4$, `print(x)`. State each assignment and the output. [4]

b) A loop starts $total = 0$ and adds i for $i = 1$ to 4 inclusive. Give the total after each iteration. [4]

c) Explain how a trace table can help find a logic error. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Complete trace tables.			
Trace selection and nested iteration.			
Use traces to locate logic errors.			